

HMITiles

ISA-101–Inspired Open-Source HMI Tile Library for Industrial Dashboards

Overview

HMITiles is an open-source HMI (Human Machine Interface) tile library written in **B4X**, following widely accepted industrial HMI design principles.

It provides reusable, professional-grade HMI tiles for: Industrial dashboards, SCADA (Supervisory Control and Data Acquisition) front-ends, Machine and process HMIs

The focus of this library is **clarity, consistency, and disciplined HMI design** - not visual effects or UI decoration.

Design Goals

- Based on widely accepted industrial HMI design principles
- Alarm-first color discipline
- Clear and consistent state handling
- Touch-friendly layouts
- No animations or visual noise
- Minimal configuration required

This is a **non-commercial project** shared for inspiration.

B4J Example – Overview HMITiles

HMITiles Example Overview v20251231 (B4J)

Label Normal Label Warning Label Alarm 11:54:10 Button Toggle Button Alarm Button OnOff

Sensor 85°C Warning Value Motion Detected SeekBar 85 Indicator 85 Gauge 85

RGB Hor RGB Vert Level 85 % Sensor 67 Trend Image

Setpoint 20 Setpoint 2 Select Item1 Item 2 Item 3 Select ON Label Half

Event Viewer

- 11:54:04 - [TileSeekBar_ValueChanged] value=85
- 11:53:52 - [TileSelect_ValueChanged] value=ON
- 11:53:52 - [TileSelect_ValueChanged] value=ON
- 11:53:52 - [TileSPPV1_SetPointChanged] value=25.0

List

- item 0
- 0
- item 1
- 1
- item 2
- 2
- item 10
- 10

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HMITiles Used
 HMITileLabel (3)
 HMITileDigitalClock
 HMITileClock
 HMITileButton (3)

HMITileSensor (3)
 HMITileSeekBar (2)
 HMITileGauge

HMITileRGB (2)
 HMITileLevel
 HMITileReadOut
 HMITileTrend
 HMITileImage

HMITileSetPoint
 HMITileSelectList
 HMITileSelect
 HMITileShape (8)
 HMITileImageIcon

HMITileEventViewer
 HMITileList

B4J Example - Water Tank Simulator

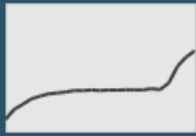
A simple interactive water tank PID simulator demonstrating ISA-101-style HMI design using HMITiles.

HMITiles Example Water Tank Simulator v20251231

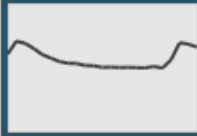
Input

FIC-101
38.2
- 45 +
Δ -6.8

Tank Level

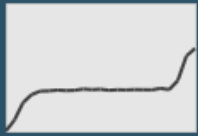
FR-101


LIC-101
67
50
Δ 17

LR-101


Output

FI-102
39.3

FR-102


Simulator
ON

VPI-102
42
Valve %


VPR-102


Event Viewer

 11:56:14 - [SimulatorStep] Input=38.2, Level=67, Output=39.3

 11:56:14 - [TileSPPVTankLevel_ValueChanged] value=67.0

 11:56:12 - [SimulatorStep] Input=35.2, Level=69, Output=35.7

 11:56:12 - [TileSPPVTankLevel_ValueChanged] value=69.0

 11:56:10 - [SimulatorStep] Input=30.8, Level=70, Output=24.1

 11:56:10 - [TileSPPVTankLevel_ValueChanged] value=70.0



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HMITiles Used

HMITileLabel

HMITileSetpoint

HMITileTrend

HMITileSetpoint

HMITileReadOut

HMITileTrend

HMITileButton

HMITileSeekBar

HMITileTrend

HMITileEventViewer

1/1/2026

Project HMITiles

3

B4J Example – ArduinoLED

Control the state of an LED connected to an Arduino UNO via the serial line.

HMITiles Used

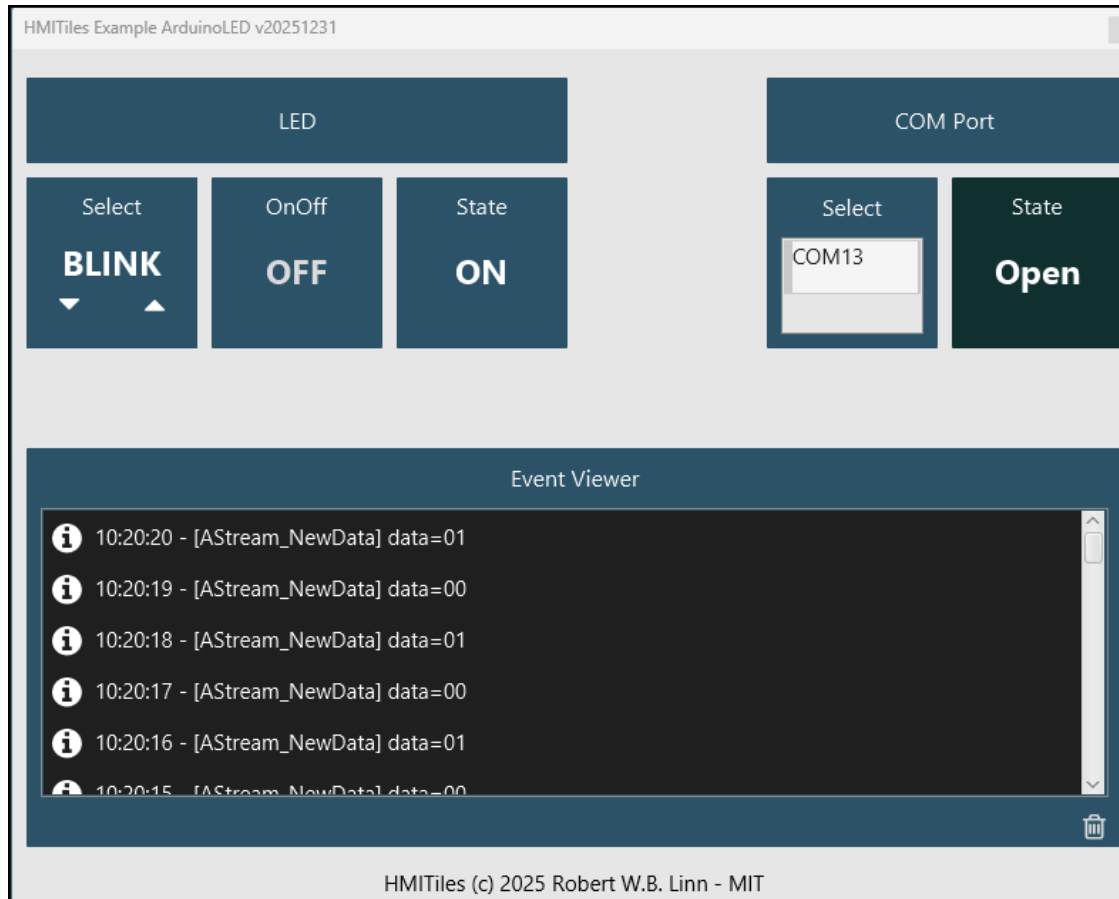
HMITileLabel

HMITileSelect

HMITileButton

HMITileReadOut

HMITileEventViewer



HMITiles Used

HMITileLabel

HMITileSelectList

HMITileButton

HMITileEventViewer

B4A Example – Overview



HMITiles Used

HMITileLabel (3)
HMITileDigitalClock
HMITileClock

HMITileButton (3)
HMITileTrend
HMITileGauge

HMITileSensor
HMITileReadOut
HMITileLevel

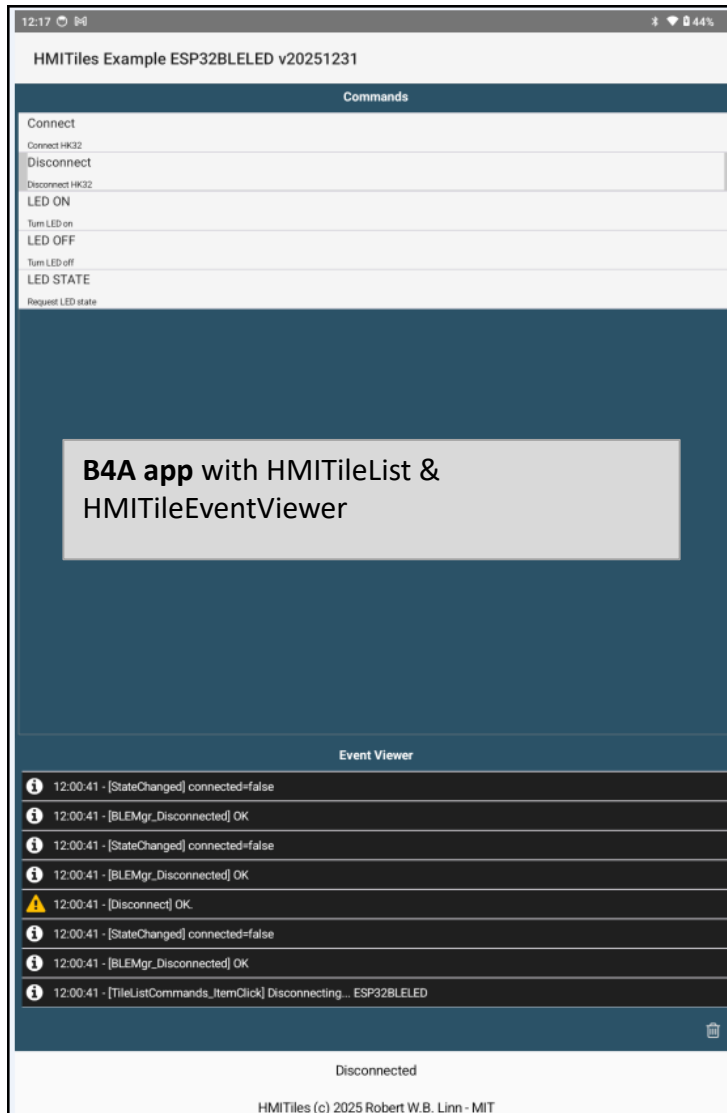
HMITileSeekBar (2)

HMITileSetPoint
HMITileSelectList
HMITileSelect

HMITileEventViewer

B4A Example – ESP32BLELED

Control the state of an LED connected to an ESP32 via the BLE (UART).

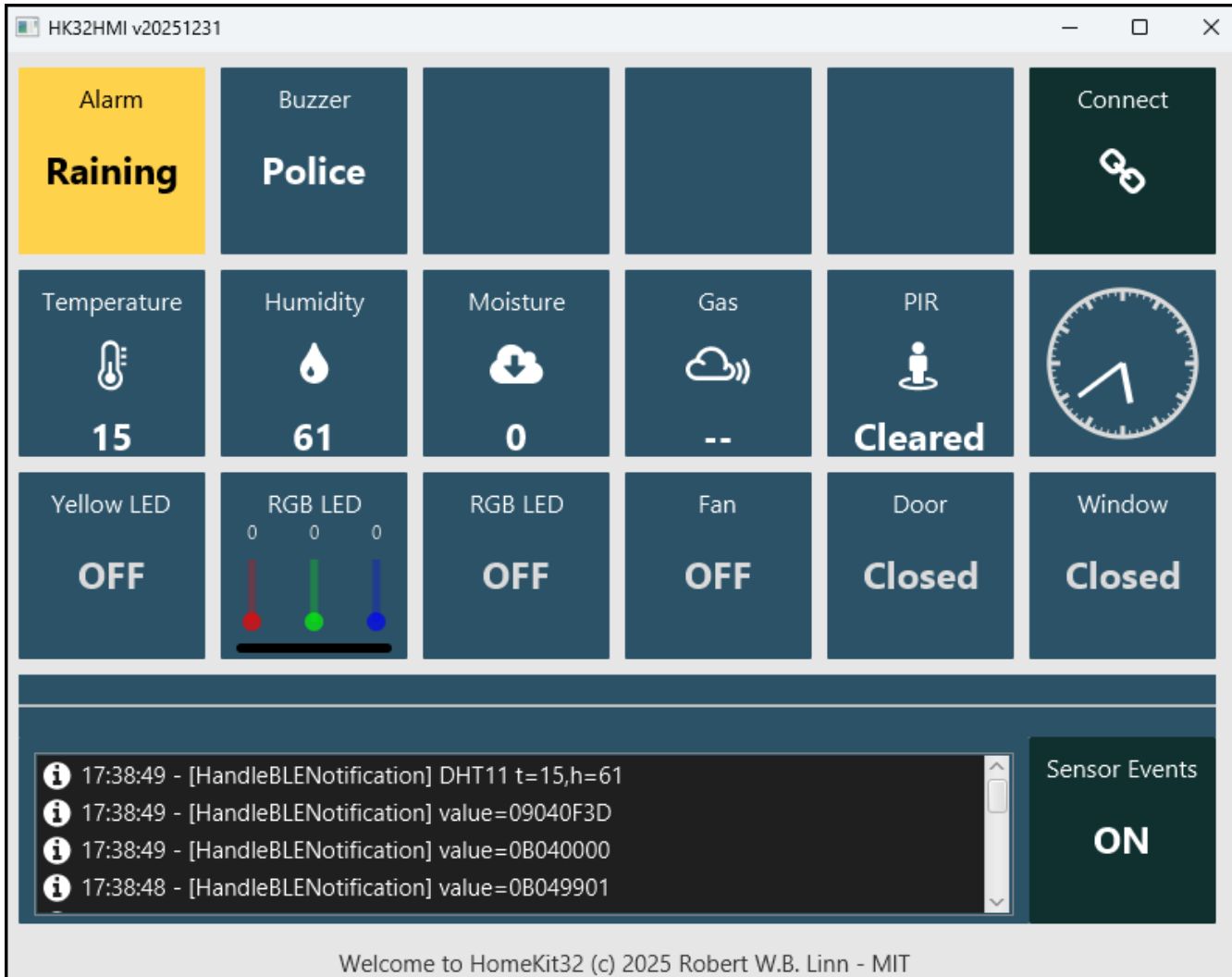


```
[B4RBLEServer::onConnect] Client connected
[CommBLE.BLEServer_NewData] buffer HEX=010101
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010101
[CommBLE.BLEServer_Write] data=010101
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010101
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=010100
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=010100
[CommBLE.BLEServer_Write] data=010100
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010100
[B4RBLEServer::onWrite] Received bytes: 3
[CommBLE.BLEServer_NewData] buffer HEX=0102
[CommBLE.BLEDispatch] deviceid=1
[DevLED.ProcessBLE] payload=0102
[CommBLE.BLEServer_Write] data=010200
[B4RBLEServer::Write] Notified bytes: 3
[DevLED.WriteToBLE] payload=010200
[B4RBLEServer::onWrite] Received bytes: 2
[B4RBLEServer::SetStartAdvertising] Started advertising: ESP32BLELED
[B4RBLEServer::onDisconnect] Client disconnected, restarting advertising
```

B4R program IDE log showing the commands:
connect > command LED ON > command LED OFF > disconnect

B4J Example – HomeKit32

HomeKit32 is a modular smart-home automation framework for the Keyestudio Smart Home Kit (KS5009), powered by **ESP32, BLE, MQTT and B4X**.



HMITiles Used

HMITileLabel (2)
HMITileButton

HMITileSensor (4)
HMITileDigitalClock
HMITileClock

HMITileButton (5)
HMITileRGB

HMITileEventViewer
HMITileButton



Screenshots